

Investigation of vortex-induced vibration of a cable-stayed bridge without backstays based on wind tunnel tests

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ABSTRACT

Cable-stayed bridges without backstays located in the abrupt valley are sensitive to wind-induced structural vibrations. The understanding of vortex-induced vibration for specific aerodynamic configuration with glass deck is of great importance for structure design. This paper thus aims to investigate the vortex-induced vibration of a cable-stayed bridges without backstays through the segmental model test and scaled full-aeroelastic model wind tunnel test. The results indicate that the bridge presents the vertical vortex-induced vibrations at the low wind speed whereas the maximum wind velocity causing the torsional vortex-induced vibrations increases to 41 m/s, which is much larger than that of the traditional bridge. The vertical and torsional vortex-induced vibrations in the smooth flow are observed at the positive wind angle of attack, owing to the responses for the bridge deck present a sharp single-frequency amplitude when the frequency of vortex-shedding is close to the flexural vibration frequency. The turbulence can effectively suppress the vortex-induced vibrations and only some small peaks are visible in the turbulent flow. The vortex-induced response of the bridge deck is significantly alleviated with increasing of damping ratio. Finally, the vortex-induced vibrations in the uniform flow disappear and the wind-induced responses are impaired with the ventilated railing in place of the airtight railing because the flow can go through the railing. It can provide a designing basis for engineering reference of cable-stayed bridges without backstays.

1. Introduction

Sightseeing glass bridges located in steep valley area are being built on a large scale owing to the development of tourism, for instance, the glass bridge located in Zhangjiajie. Consequently, their aerodynamic performance is seriously affected by the wind field characteristics created by strong wind and special terrain, such as wind speed, wind angle of attack, and complex aerodynamic configuration. These bridges are more susceptible to vortex-induced vibration (VIV) than traditional bridges due to the complex fluid–structure interaction and light mass. Therefore, investigation of the VIV and accurate predictions of vortex-induced resonance response of the bridges become important and necessary to ensure pedestrian comfort and safety.

Numerous studies report that VIV is a typical vibration induced by the fluid–structure interaction [1–4]. Vortex shedding occurs with flow passing through the vibrating bridge deck, especially, the wind direction

is nearly perpendicular to the bridge axis in a low turbulence intensity, leading to enhance the vortex shedding [5,6]. Yang et al (2015), Park and Kim (2017), and Rizzo et al (2018) mainly focus on the cable-stayed bridge, indicating that three dimensional properties of vortex shedding around the cable play an important role in the VIV's mechanisms [7–9]. Note that the vortex-induced resonance will occur if the frequency of vortex-shedding closes to the flexural vibration frequency of the bridges [10–13]. Although, VIV is ascribed to a restricted vibration, it may cause fatigue damage to structural components of bridge and impair the pedestrian safety. Hence, accurate prediction of vortex-induced response (VIR) should be performed in designing the deck shape of bridges.

Many researchers have developed different models to predict the locked-in vibration of the bridge deck. Ehsan (1990) predicted the responses of several long-span bridges by the Scanlan–Simiu model, the results were not validated with full-scale measurements [1]. Wu and

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Kareem (2013) investigated the VIV of bridge decks using the Van der Pol-type equation, the results indicated that the increase of Scruton number leads to the sudden change of stability of the VIV system at the end part of lock-in wind velocity region, and the differences in structural detail of similar cross-sectional shapes may strongly affect the vortex-induced response of a body [14]. Obviously, the existing empirical models can not properly predict VIV response due to the influence caused by the considerable differences of the aerodynamic configuration and cross-section of bridge on the characteristics of locked-in vibrations [15–17,19]. Thereby, the wind tunnel test becomes a standard tool in bridge design. The present study experimentally investigated the VIV phenomenon of a CSBB with D-section girder.

The sophisticated aerodynamic configuration of cable-stayed bridges without backstays (CSBB) also contributes to the inaccurately quantifying the wind effects on the structure. On the other hand, the research object is a typical bluff body structure, whose main girder cross-section is the D-section. Zhao et al (2018) have contributed to the importance of after-body to system on vortex-induced vibration, and new findings show that the D-section behaves a VIV-dominated response for the reduced velocity <10 and an afterbody is not essential for VIV at low mass and damping ratios [18]. VIVs for many intricate bridges in practical engineering have been widely investigated, such as box-shaped bridges, long-span cable-stayed bridges, and cantilever bridges [7,8,19,20]. Consequently, many points on wind-induced vibration of bluff bodies remain unclear. Besides, most current wind codes (ASCE7-10, JTG/T 3360-01-2018, EN 1991-1-4:2005) also do not provide relevant guidelines for determining the wind effects on such type of bridges [21–23]. Therefore, considering the significant differences for the vortex structures and VIVs induced by aerodynamic configurations, this paper studied the VIVs for a CSBB with extraordinary geometry through wind tunnel test.

VIVs often happen in a relatively low wind speed range about 6–10 m/s, leading to suffer from the long-term vibration [20,24]. Chen et al (2014) and Zhu et al (2018) reported that the torsional VIV is prone to in the uniform flow at a positive wind angle of attack, and the VIV disappears in the turbulent flow for long-span cable-stayed bridges, and cantilever bridges [24,25]. It is evident that wind speed, wind angle of attack, and flow-field characteristics will all induce the VIVs. Hence, the influence of aforementioned factors on the VIVs for a CSBB is systematically studied. It is inferred from previous studies that the aerodynamic response decreases with increasing of structural damping or adjusting aerodynamic configuration [10]. Larsen et al (2000), Bai et al (2020), Lupi et al (2018), and Patil et al (2011) proposed an alternative aerodynamic mitigation measure to alleviate the VIV for bridges and verified its efficacy through wind tunnel tests [3,26–28]. In another word, introducing a reasonable aerodynamic configuration contributes to impair the VIV amplitude or makes the VIVs disappear to ensure users' comfort and improve the life of bridges.

This paper reports the wind tunnel test measurements of aeroelastic properties of a CSBB with a group of scaled full-aeroelastic models. In the wind tunnel test, 11 cases are designed to investigate the influence of flow-field characteristics, wind angle of attack, and wind speed on the aerodynamic response for the CSBB. The VIVs are analysed based on the statistics of vertical displacement and acceleration, lateral displacement, and torsional angle. Besides, this study investigates the effect of the damping ratio on the aerodynamic response of bridges and proposes a reasonable aerodynamic configuration to improve the aerodynamic performance of the bridge.

2. Description of bridge

A CSBB consists of a 60-m-long cantilever girder, reinforced concrete arch towers, and plexiglass deck. The lower arch tower embedded in the mountain is inclined at 45° , and the other one embedded in the mountain with pile and anchor is designed with 30° inclination angle. Nine stay cables, made of steel, are mounted on each side, and the stay cables

fixed on the arch tower have 3.1 m horizontal spacing, while those connected to the main girder are set to have 5.2 m horizontal spacing, as displayed in Fig. 1. The main girders are made of a steel pipe with 110 cm diameter, as illustrated in Fig. 1(c), and the supporting system adopts the cantilever structure system comprising an arch, girder, and consolidation foundation.

The cantilever length of the bridge increases with the demand for sightseeing and the glass deck is built to enrich the sightseeing experience, and such requirements normally leads to the characteristics of lighter mass, lower natural frequency, and smaller damping of the bridge. As a result, the CSBBs are becoming more sensitive to wind loads than traditional bridges, besides, the sophisticated geometric shape of the bridge also contributes to the inaccurately quantifying the wind effects on the structure. The bridge would suffer from stronger gusty wind due to the impact of sophisticated terrain, and hence the bridge is susceptible to vortex-induced vibrations.

3. Frequency analysis of CSBB

To confirm that the dynamic characteristics of full-aeroelastic model meet the wind tunnel test requirements, the frequency analysis of the cable-stayed bridge was firstly performed using the commercial software ANSYS before the wind tunnel test. Both the main girder and pylon adopted the spatial beam element to reach the computational accuracy, and the stay cables were simulated with the link element transmitting only tension forces.

3.1. Finite-element model

The FE model of the bridge was developed by the commercial software ANSYS to investigate its dynamic characteristic parameters, as illustrated in Fig. 2. Both the main girder and pylon were simulated by the element BEAM 188, where the pylon was composed of a variable cross-section with a maximum dimension of $1.2 \text{ m} \times 2.93 \text{ m}$ and an equal dimension of $1.2 \text{ m} \times 1.2 \text{ m}$. The prestress force among the first seven cables behaves slight difference by analysing the relationship between the horizontal tension and sagging before creating the FE model. Thereby the nine steel stay cables were mounted on each side and simulated by the element LINK10 with a 150 kN tension force; however, the tension force of the eighth and ninth stay cables have the values of 175 kN and 200 kN, respectively, whose numbers are shown in Fig. 1(a).

It is noticeable that the design values of cable tension are relatively small due to the utilization of glass bridge deck. For the finite element model, the elastic modulus of the stay cable is modified using the Eq. (1) [29] owing to it being modelled with truss elements, such that it can reflect the sagging occurrence in the cables.

$$E_r = \frac{E_0}{1 + E_0 \frac{(\rho_c g l \cos \alpha_c)^2}{12 \sigma^2}} \quad (1)$$

where E_r is the elasticity modulus of the stay cable after considering the sagging effect; E_0 is its original elasticity modulus; ρ_c represents its density; l is its length; α_c is its inclination angle; σ is average stress, and g is the gravity acceleration.

The arch towers are fixed in the weathered rock, and the main girder on one end is fixed in the weathered rock and free on the other end. The arch tower and main girder are connected by nine stay cables on each side with different tension forces, and the hinged connection is adopted at their both ends. The vertical displacements of the bridge deck at the connections with the stay cable under gravity load are displayed in Fig. 3, and the bearing capacity of stay cables has been verified by Eq. (2) (JTG/T 3360-01-2018) [22] to meet the design requirements.

$$\frac{\gamma_0 N_d}{A} \leq \phi_d f_d \quad (2)$$

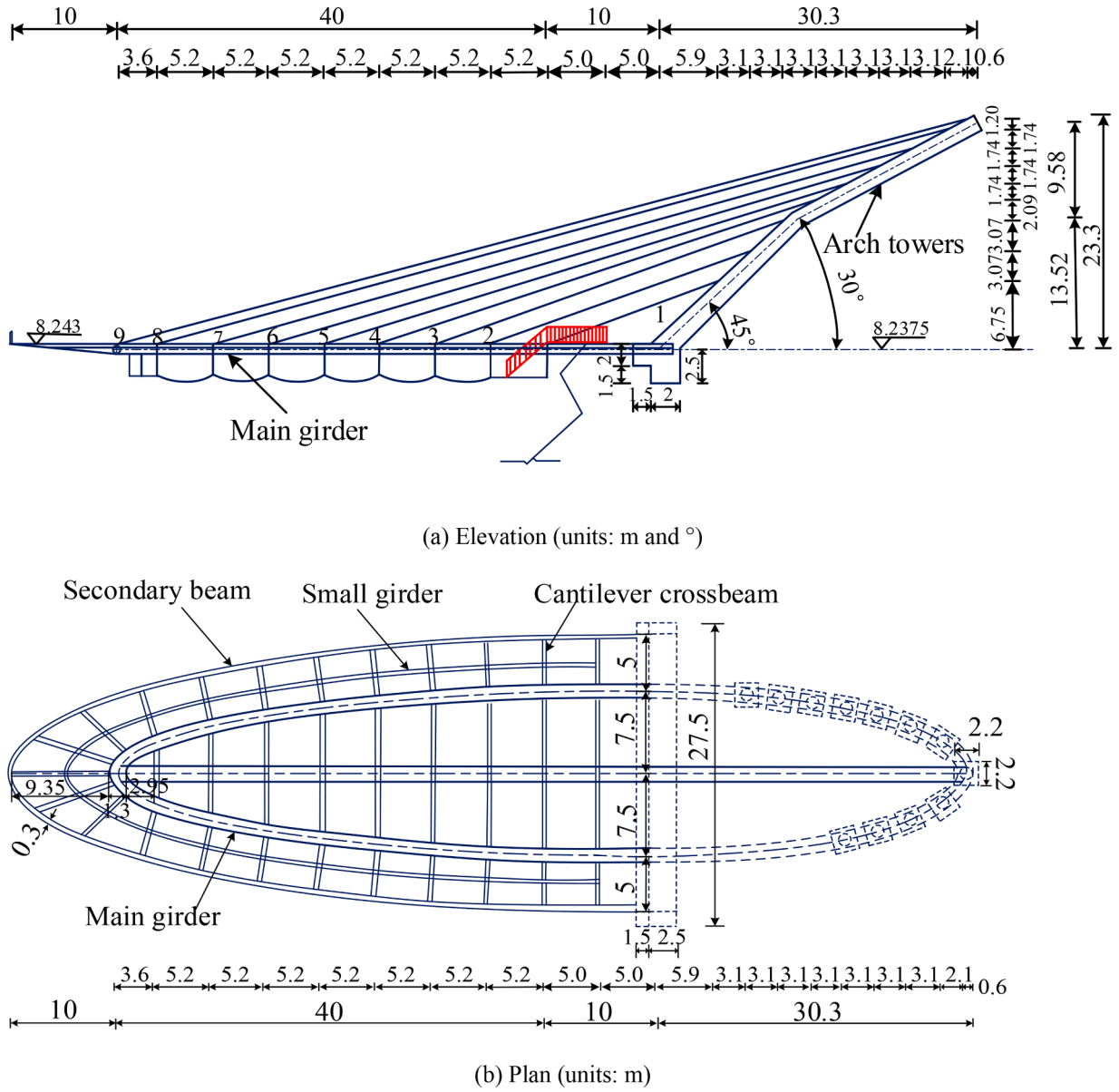


Fig. 1. Schematic diagram of bridge and girder cross-section (m and °): (a) elevation; (b) plan; (c) cross-section.

where γ_0 is the coefficient for importance of a structure taken with a value of 1; N_d is design value of axial tension of stay cable; A is the sectional area of stay cable; ϕ_d is the correction factor for stay cables taken as 1.0; f_d is the design value of tensile strength of stay cable (MPa).

3.2. Results of mode shapes

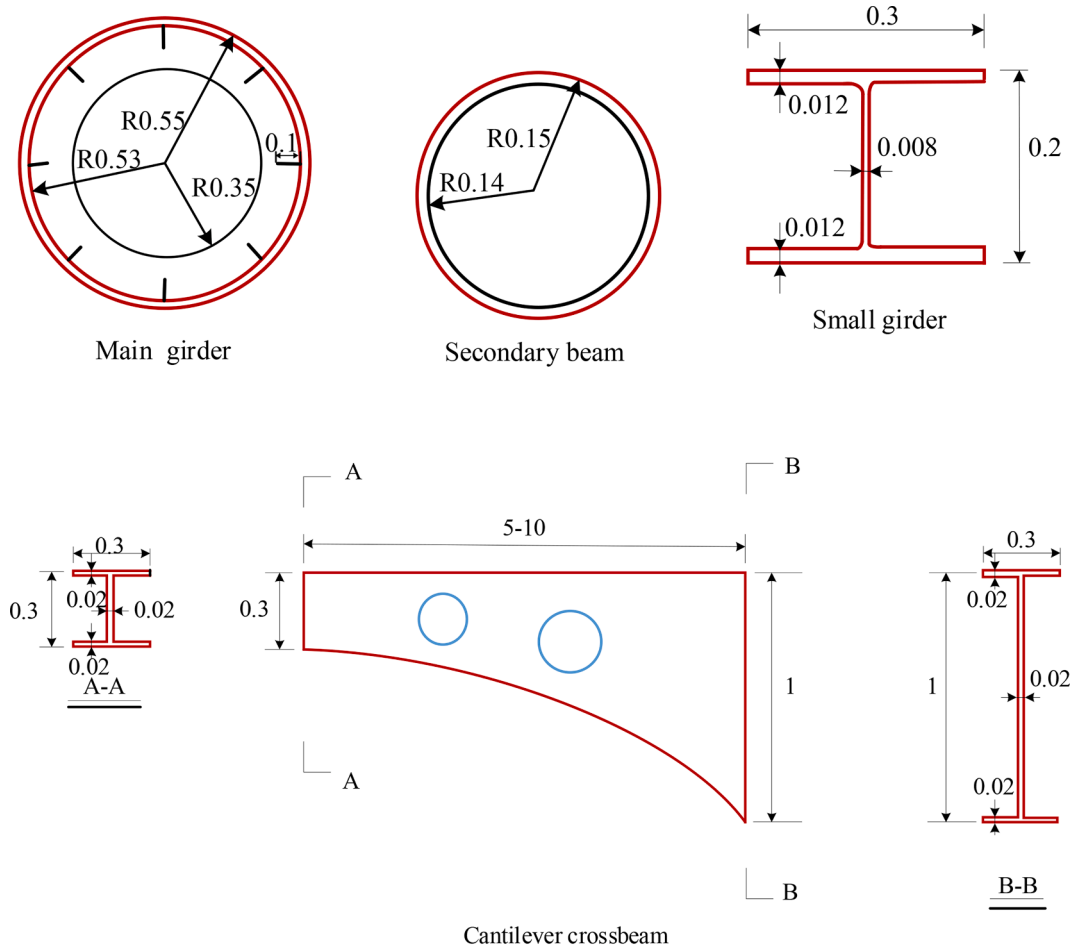
The mode shapes with low frequencies are more prone to occur than the remaining mode shapes with high frequencies due to the wind frequency, except for the situations of strong winds. To obtain the structural dynamic characteristics, a geometric nonlinear static analysis was performed before the modal analysis to account for large displacement effects subjected to dead load and the stress stiffness of the cables. The effect of internal force of the structure induced by the dead load was also considered based on the static analysis result.

Fig. 4 describes the first three modal shapes of the finite element model. The results indicate that only the first-order torsional mode is antisymmetric and the other modes are asymmetric for the first nine

modes. The first three mode shapes are vertical and lateral bending, and torsion and their corresponding natural frequencies are 0.366 Hz, 0.614 Hz, and 0.921 Hz, as listed in Table 1. There are significant coupling effects between the second-order lateral bending mode and torsional mode.

4. Full-aeroelastic model test procedure

To measure the wind-induced response of the bridge, a full-aeroelastic model was created based on the analytical results and similarity criterion to obtain the aerodynamic performance of the bridge. The wind-induced vibration responses of the main girder in uniform and turbulent flow, as well as the damping ratio effect on the aerodynamic response, were investigated. An aerodynamic configuration improvement measure was proposed to enhance the aerodynamic stability and mitigate the vibration amplitude.



(c) Girders' cross-section (units: m)

Fig. 1. (continued).

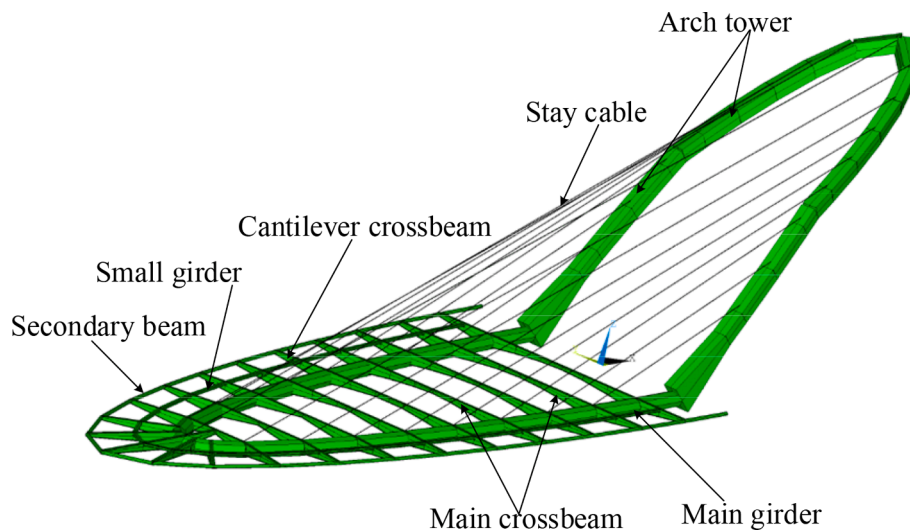


Fig. 2. Finite-element model of the bridge.

4.1. Wind speed determination

According to the JTG/T 3360-01-2018 code [22], terrain characteristics of the valley area, and the measured wind speed by cobra probe

at the bridge site, the basic wind speed was determined as 44.5 m/s. The variation of wind speed along the bridge span indicates that the spatial distribution characteristics should be considered to simulate the wind loads because the wind speed distribution along the bridge span is

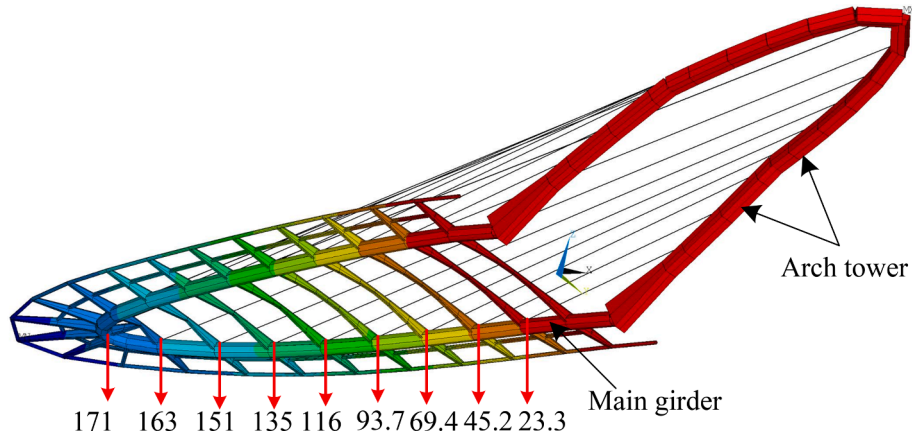


Fig. 3. Vertical displacements of the deck at the connections with the cables (unit: mm).

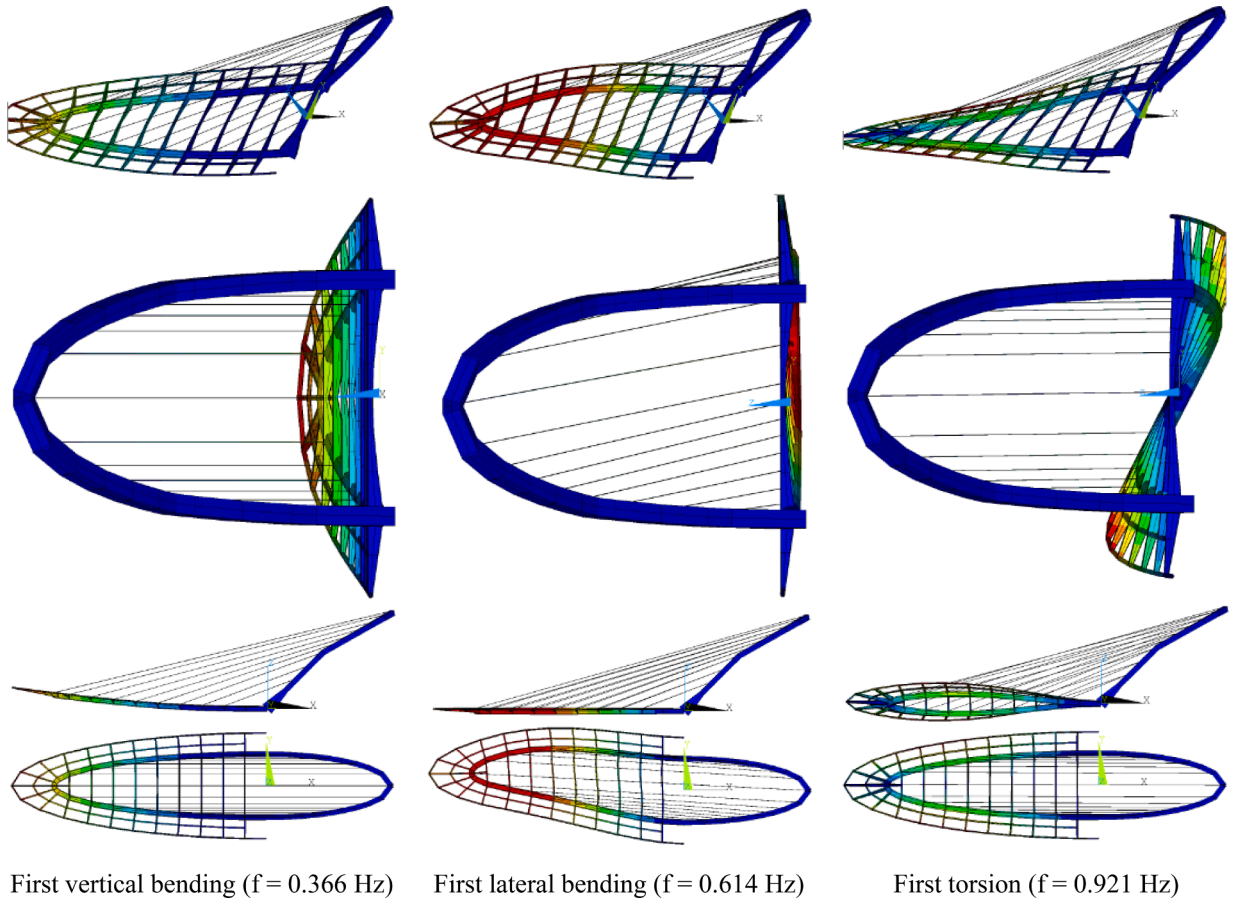


Fig. 4. First three modes shape of bridge.

affected by the mountain's shape. The acceleration effect of valley wind should consider the influence of the specific topography of the hillsides on both sides and the position and height of the measuring point [27,28].

The wind speed change rate increases with increasing measurement point height; thus, the equivalent height method is adopted in this study because the bridge deck height varies substantially along the bridge span due to the relief terrain in the valley. To obtain the variation characteristics of the wind speed at the valley and bridge deck, the benchmark wind speed is determined based on the average height 166.6 m at $1/2L$, $3/4L$, and L , where L represents the bridge length, and the corresponding vertical distances are 30 m, 150 m, and 320 m. The bridge

site benchmark wind speed of the main girder is determined as 58.4 m/s obtained from Eq. (3) (JTG/T 3360-01-2018) [22].

$$V_d = 0.564 \times V_{10} \times \left(\frac{Z}{10}\right)^{\alpha} \quad (3)$$

where V_d represents the bridge benchmark wind speed; V_{10} is the basic wind speed; Z is the average height at the bridge deck, and α is the exponential constant. Considering the terrain characteristics of the surrounding environment, the bridge site is regarded as category D stipulated in the JTG/T 3360-01-2018 code and the constant α is set as 0.3 [22]. The detailed parameters to determine the wind speed were

Table 1
Modal parameters of the bridge.

Number	Computed prototype frequencies (Hz)	Modal mass (ton)	Mode
1	0.366	79.401	V-1
2	0.614	127.504	L-1
3	0.921	14631.217	T-1
4	1.180	109.869	V-2
5	2.531	1685.015	T-2
6	2.736	89.475	L-2
7	2.878	131.278	V-3
8	4.478	4909.374	T-3
9	4.892	208.908	V-4

Note: V, L, and T, respectively, represent the vertical and lateral bending, and torsion of the deck; number represents the mode order.

summarized in Table 2.

4.2. Measurement solution

The measuring points were mounted on the main girders to obtain the wind-induced vibration response of the main girders, presented in Fig. 5(a) and Fig. 5(b). In this study, 15 loading conditions, as listed in Table 3, were investigated, including various wind angles of attack (α) ranging from -3° to $+3^\circ$ with an increment of 3° , wind yaw angles (β) with an increment of 15° (shown in Fig. 5(c)) in the uniform and turbulent flows, different railing schemes, and damping ratios. The turbulent intensity was about 2% in the uniform flow without wind speed gradient in the vertical direction; while the turbulent flow was simulated as category D in this test.

4.3. Full-bridge aeroelastic model design

(1) Similarity criterion and simulation

The similarity of geometrical shape and size is required to be matched to accurately evaluate the aerodynamic performance during the process of designing and manufacturing a scaled full-aeroelastic model. Accordingly, a scaled full-aeroelastic model focusing on six similarities—geometric shape, Froude's number, density ratio, Cauchy number, Strouhal number, damping ratio, etc. [21,22], was made by acrylonitrile-butadiene-styrene plate. The geometric scale ratio of the full-aeroelastic model was determined as 1/29 due to the bridge size and the size effect of the high-speed working section of the HD-2 wind tunnel. The dimensions of working section of the HD-2 wind tunnel are 17 m (length) $\times$ 3 m (width) $\times$ 2.5 m (height). The research object in the wind tunnel test is a typical bluff body structure, and the maximum Reynolds number is distributed in the range of $2.24 \times 10^5 \sim 4.48 \times 10^5$. Consequently, the inertia effect plays a key role, resulting in flow separation, and the Reynolds number was neglected owing to its slight effect on a typical bluff-body structure during the wind tunnel tests.

To prevent the bridge's shape from controlling the structural stiffness, the main girder was divided into several sections with a gap of 2 mm between each section. The steel installed at the two adjacent segments were used as the skeleton for achieving the stiffness of the full-aeroelastic model, and the size of steel skeleton was then determined according to the similarity of bending and torsional stiffness. Considering the similarity of axial stiffness related to the geometric scale (n)

Table 2
Parameters of design wind speed.

Category	Value	Description
Basic wind speed (m/s)	44.5	10-m height, 10-min interval, 100-year return period
Coefficient of wind profile	D, 0.30	Exponent
Design standard wind speed (m/s)	58.4	/

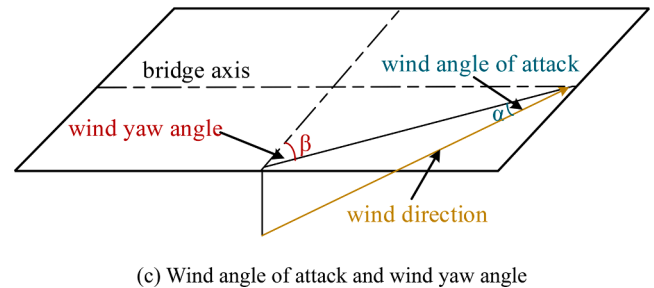
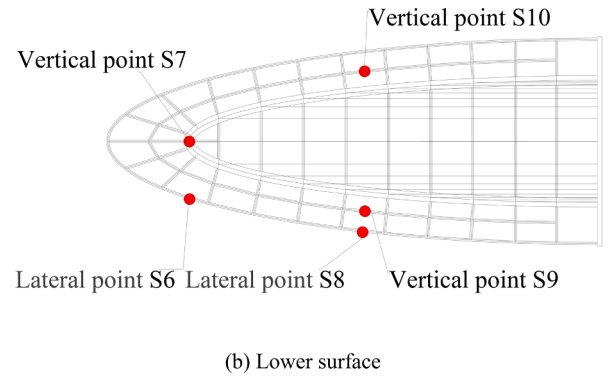
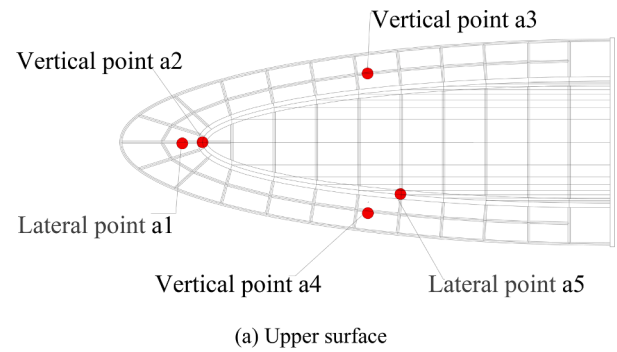


Fig. 5. Arrangement of measuring points, for acceleration (a in Fig. 5(a)) and displacement (S in Fig. 5(b)) for the selected channels, and wind angle of attack and wind yaw angle (in Fig. 5(c)).

and wind speed ratio (m), a tension spring conforming to the axial stiffness was designed, while the other cables adopt a thin steel wire with larger axial stiffness. Finally, the full-aeroelastic model also needs to consider the structural mass system to ensure the similarity of the structural dynamic characteristics, and the materials with higher hardness and less deformation energy consumption were selected to reduce the material damping.

(2) Model similarity ratio

According to the aforementioned similarity criteria and bridge size, the geometric scale of the full-aeroelastic model in the wind tunnel test was determined as 1/29, and thus, the blockage ratio was $<5\%$. This tiny error is generally accepted in wind engineering. The main similarity parameters of the full-aeroelastic model were obtained based on the geometric scale ratio, listed in Table 4.

The main girder in the laboratory model is arranged horizontally with the same as in FEM model (Fig. 1(a), Fig. 3 and Fig. 6). The two arch towers are both embedded in the mountain, and the fixed connection is adopted between the arch tower and main girder, such that the arch towers are designed to be vertical in the laboratory model, as shown in Fig. 6. There is difference only in the railing design for the two full-aeroelastic models. Fig. 6(a) depicts the model with airtight railing,

Table 3

Test case.

Case	Structural state	Flow field	Wind angle of attack	Wind yaw angle
DH-1	A	Uniform flow	0°	0°
DH-2	A	Uniform flow	0°	15°
DH-3	A	Uniform flow	0°	30°
DH-4	A	Uniform flow	0°	45°
DH-5	A	Uniform flow	0°	60°
DH-6	A	Uniform flow	0°	90°
DH-7	A	Uniform flow	+3°	0°
DH-8	A	Uniform flow	−3°	0°
DH-9	A	Turbulent flow	+3°	0°
DH-10	A	Turbulent flow	0°	0°
DH-11	A	Turbulent flow	−3°	0°
DH-12	B	Turbulent flow	0°	0°
DH-13	B	Uniform flow	0°	0°
DH-14	C	Uniform flow	0°	0°
DH-15	D	Uniform flow	0°	0°

Note: A represents the completed bridge, B is the airtight railing with additional damping, C denotes the ventilated railing, and D represents the ventilated railing with additional damping.

Table 4

Main similarity parameters.

Parameters	Symbol	Unit	Similarity	Similar requirements
Length	L	m	$\lambda_L = 1:29$	Geometric similarity
Velocity	U	m/s	$\lambda_v = \sqrt{\lambda_L} = 1:5.385$	Froude
Acceleration	g	m/s ²	$\lambda_g = 1$	\
Density	ρ	kg/m ³	$\lambda_\rho = 1$	\
Time	t	s	$\lambda_t = \lambda_L/\lambda_v = 1:5.385$	Strouhal
Frequency	f	Hz	$\lambda_f = \lambda_v/\lambda_L = 5.385:1$	Strouhal
Bending stiffness	EI	N•m ²	$\lambda_{EI} = \lambda_L^5 = 1:29^5$	\
Torsional stiffness	GJ _d	N•m ²	$\lambda_{GJ} = \lambda_L^5 = 1:29^5$	\
Axial stiffness	EA	N	$\lambda_{EA} = \lambda_L^3 = 1:29^3$	\
Mass	m	kg/m	$\lambda_m = \lambda_\rho \lambda_L^2 = \lambda_L^2 = 1:29^2$	\
Inertia moment	I _m	kg•m ² /m	$\lambda_J = \lambda_m \lambda_L^2 = \lambda_L^4 = 1:29^4$	\
Damping ratio	ξ	–	$\lambda_\xi = 1$	\

while the other model's railing is designed as a ventilated railing (see in Fig. 6(b)). The ventilated railing's height is 2.05 m as same as the airtight railing's height, whose spacing is set to 0.1 m.

(3) Dynamic characteristics of full-aeroelastic model

The dynamic characteristics focus on the first few modes that are more sensitive to wind loads. In the design of a scaled full-aeroelastic model, the first-order lateral, vertical, and torsional symmetrical and asymmetrical modes are often chosen to satisfy the similarity criterion [30]. To validate the dynamic characteristics of the full-aeroelastic model, its modal parameters were compared with the corresponding modal parameters of the finite element model.

To obtain the first three modes shape of the full-aeroelastic model, the bridge was simplified to a cantilever beam because the fixed

connection is adopted between the arch tower and main girder, and free at the other end is. Meanwhile, the modes shape of the finite element model was obtained by the normalization method based on the nodal displacement at the same position as the test points. Then the first three modes shape of the full-aeroelastic model were obtained based on the displacement response and compared with the modal shapes of the finite element model to validate the tests (see in Fig. 7). The results indicate that the first three modal shapes between the finite element model and the full-aeroelastic model show a good agreement.

The damping ratios and the natural frequencies of the full-aeroelastic model are respectively identified by the half-power bandwidth method (Eq. (4)) and the Fourier analysis method (FAM) [31] (listed in Table 5) based on the acceleration response. The time-history accelerations at the measuring Points 2 and 5 were obtained by artificial excitation, as illustrated in Fig. 8 and Fig. 9. The artificial excitation is a modal impact hammer, which effectively combine the force sensor with the striking surface of the hammer. The impact hammer hammered the test model, and then the force sensor recorded the exciting force and the corresponding response of the measuring point in the modal test process. To obtain the frequency response curve and the modal parameter, the frequency response functions of the measuring point were calculated according to the exciting force on the hammer and the response data measured by the sensor.

$$\xi = \frac{f_2 - f_1}{2f} \times 100\% \quad (4)$$

where ξ is damping ratio; f_1 and f_2 are the abscissa value of the two intersection points of 0.707 times to the resonance peak in the power spectrum curve; f is the measured resonance frequency on the spectrum curve, namely, the natural frequency.

The table presents that the relative error between the measured and numerical frequencies is controlled within 3%. It is evident that there are good agreements between the measured and numerical frequencies. The damping ratio (about 0.73%) of the first vertical bending mode also conforms to the test requirements, and hence the results indicate that the scaled full-aeroelastic model satisfies the accuracy requirements of the wind tunnel test.

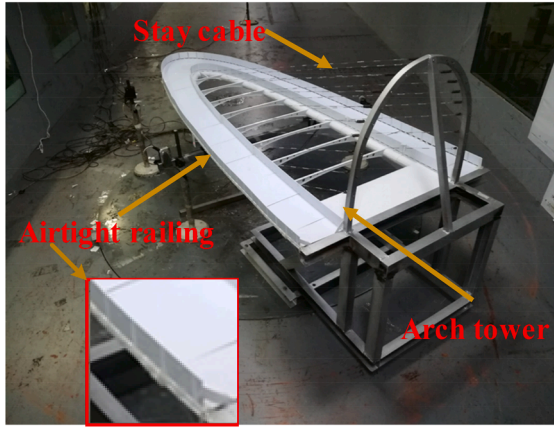
4.4. Test equipment and instruments

The wind tunnel experiment for the cable-stayed bridge was performed at a boundary-layer wind tunnel laboratory with three working sections. The full-aeroelastic model testing was conducted in a high-speed working section, and the wind tunnel dimensions are 17 m (length) × 3 m (width) × 2.5 m (height) with the loading capacity of wind speed ranging from 0 m/s to 58 m/s.

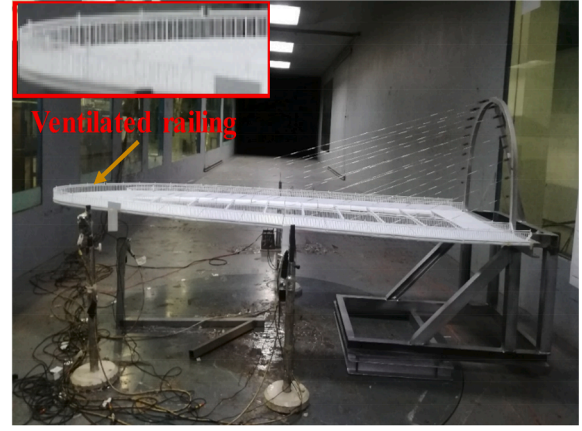
During the wind tunnel test, the wind speed and wind yaw angle around the main girder were measured using a cobra probe with 155 mm length and maximum 2.6 mm width. The sensitivity of wind speed and wind yaw angle reach 0.3 m/s and 1°, respectively. To obtain displacement and acceleration, the test also required five accelerometers and displacement meters, which were mounted at the measuring points. The measurement range and the sensitivity of the displacement meter are respectively 200 mm and 0.2 mm; Meanwhile, a DH5920 dynamic test system with specialised test software was adopted, which could complete the synchronous acquisition of displacement and acceleration in the test through 16-channel dynamic strain.

4.5. Wind field simulation

The wind field at the bridge site was regarded as the D category according to the JTG/T3360-01-2018 code [22], and the terrain characteristics of the cable-stayed bridge site; thus, the wind speed was determined as Eq. (5).



(a) Airtight railing



(b) Ventilated railing

Fig. 6. Full-bridge aeroelastic model.

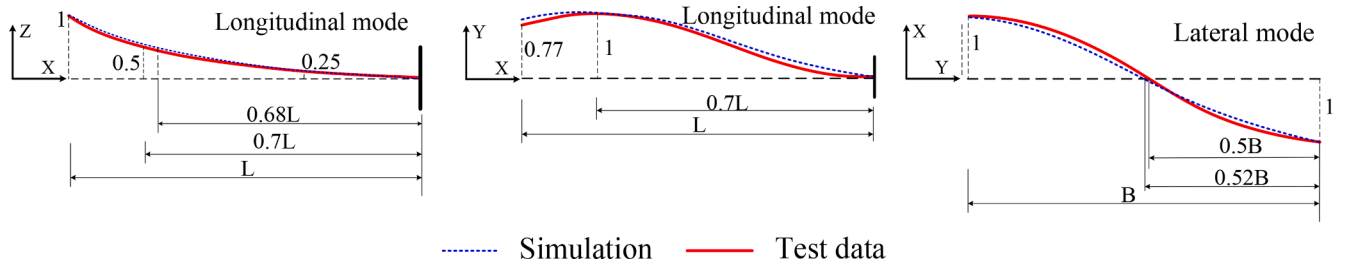


Fig. 7. First three modes shape of the finite element model and full-bridge aeroelastic model.

Table 5

Test results of dynamic characteristics.

Number	$\lambda_f \times$ Numerical frequencies (Hz)	Measured model frequency (Hz)	Error (%)	Damping ratio (%)
1	1.971	2.002	1.573	0.73
2	3.306	3.271	-1.059	0.81
3	4.96	4.858	-2.056	0.62
4	6.354	6.489	2.123	\
5	13.630	14.005	2.752	
6	14.734	15.106	2.526	

Note: λ_f the frequency similarity ratio, as listed in Table 4.

$$U(z) = U_{10} \left(\frac{z}{10} \right)^\alpha \quad (5)$$

where $U(z)$ is the mean horizontal wind speed at the height z ; z is the height from the ground; U_{10} represents the mean wind speed at 10-m height, and α is the roughness index. Considering the terrain characteristics of the surrounding environment, the bridge site is regarded as category D stipulated in the JTG/T 3360-01-2018 code and the constant α is set as 0.3 [22].

Note that the wind-induced responses of the bridge obtained in the two wind fields were that of the measuring points at the bridge deck, so no wind speed gradient behaves in the vertical direction, also the turbulence intensity gradient. Some roughness elements and spires equipped in the front part of the model were used to simulate the atmospheric boundary layer, as illustrated in Fig. 10; subsequently, a turntable located at the bottom with a rotation angle ranging from 0° to 360° was adopted to vary the applied wind direction.

In the wind tunnel test, the influence of the surrounding mountains was not considered because the elevation of the surrounding mountains in the superior wind direction is larger than that at the bridge site's, on the other hand, the wind resistant stability of the bridge could be safely

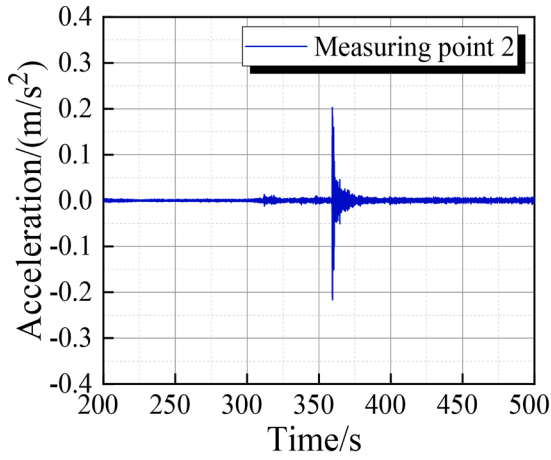
investigated without considering the shielding effect of the surrounding mountains.

The benchmark wind speed of the bridge in the completed state is 58.4 m/s, and the corresponding model test wind speed is 10.8 m/s, so the test wind speed is controlled within the range of 0 m/s to 19.5 m/s. The reference point height was set at 0.8 m in front of the main girder to ensure that the tested wind speed exhibited the actual wind speed as reaching the leading edge of the main girder. The wind field effectiveness was identified by testing the wind field characteristics, and then, the test results report that the turbulence intensity at the bridge deck in the turbulent and uniform flow are respectively 16.5% and 2%, which both are measured by the cobra probe.

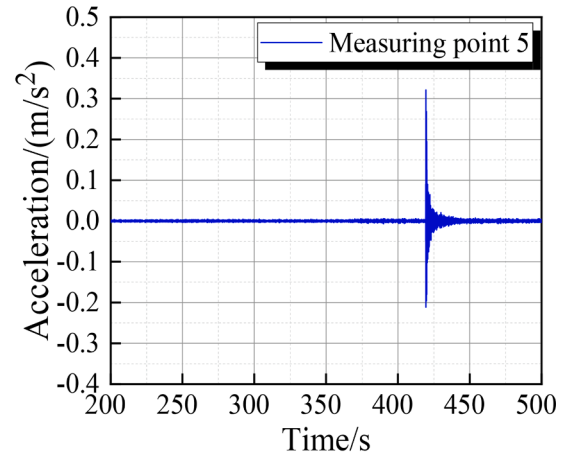
The turbulence integral scale of fluctuating wind speed can reflect the correlation of fluctuating wind speed at two points in space. However, there is difficulty to perform the simultaneous wind speed measurement at multiple points in the wind tunnel test. Hence, the wind speeds at two points are converted to the wind speed of the same point at different times based on the Taylor hypothesis. The turbulence integral scale is calculated by the auto-correlation function (Eq. (6)) at the height of bridge deck, as shown in Fig. 11. It is evident that there is no definite relationship between turbulence integral scale and wind speed. However, there are the larger turbulence integral scales in the low wind speed range, indicating that the fluctuating wind has a significant correlation in different parts of bridge, may leading to VIV.

$$L_u' = \frac{U}{\sigma_u^2} \int_0^\infty R_u(\tau) d\tau \quad (6)$$

where L_u' is the longitudinal turbulence integral scale; U is the mean wind speed; σ_u is the fluctuating wind speed RMS; τ is time interval; $R_u(\tau)$ is the auto-correlation function.



(a) Measuring point 2



(b) Measuring point 5

Fig. 8. Time-history accelerations.

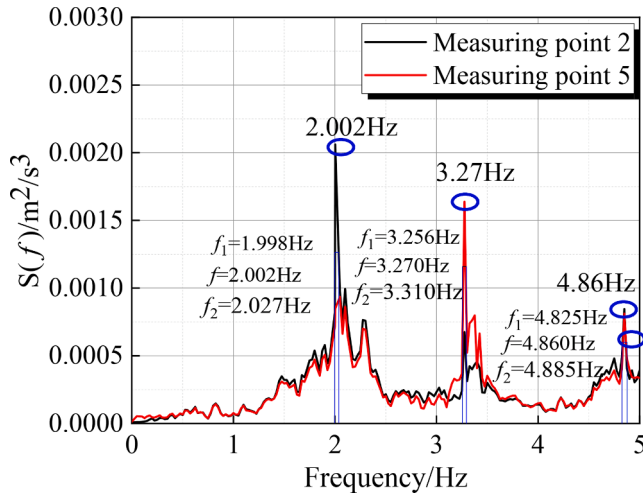


Fig. 9. Measured frequency.

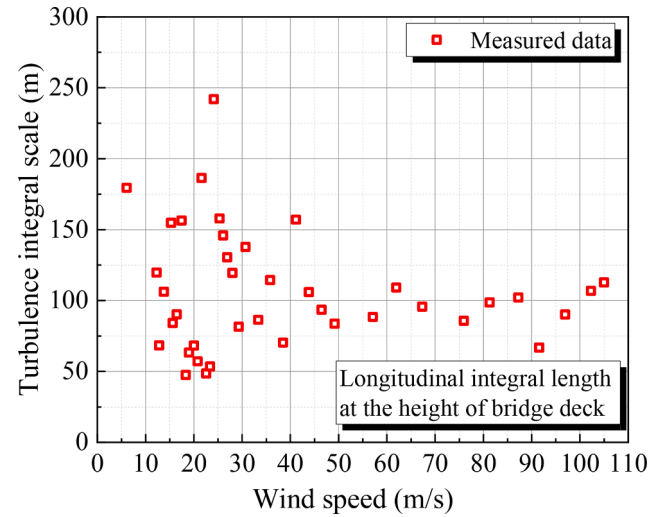


Fig. 11. Longitudinal turbulence integral scale.



Fig. 10. Wind field of wind tunnel test.

4.6. Three-component force coefficients test

The full bridge was divided into four sections during the test to obtain more accurate static three-components, and the corresponding four models (MOD1, MOD2, MOD3, and MOD4) were developed by

removing each segment, as presented in Fig. 12. The sectional model with different height was fixed vertically to the two high-frequency balances installed at the ends of the upstream and downstream of the main girder. Fig. 13 displays the reference system used for calculating the aerodynamic forces.

Twenty-five values of the wind angle of attack (α) were considered in the interval between -12° and $+12^\circ$, with increments of 1° , and the positive angles were “nose up” in relation to the approaching flow. Additionally, the wind speed in the wind tunnel test was set as 7.02 m/s to avoid lateral sloshing of the model; concurrently, an automatically controlled dial was adopted to vary the tested wind angle of attack.

The average values of the drag coefficient, lift coefficient, and torsional moment coefficients per unit span—i.e. C_D , C_L , and C_M , respectively, in the wind coordinate system of the cross section for the main girder—were evaluated according to Eq. (7-1)–(7-3) [19]. The corresponding graphs as functions of the wind angle of attack are illustrated in Fig. 14.

$$\text{Drag : } C_D = \frac{F_D}{\frac{1}{2}\rho U_\infty^2 H} \quad (7-1)$$

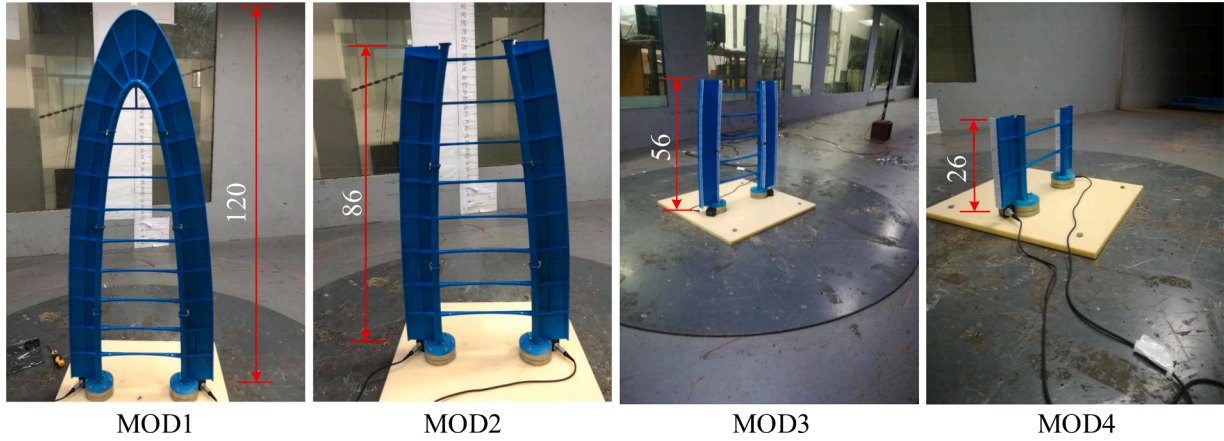


Fig. 12. Sectional model (unit: cm).

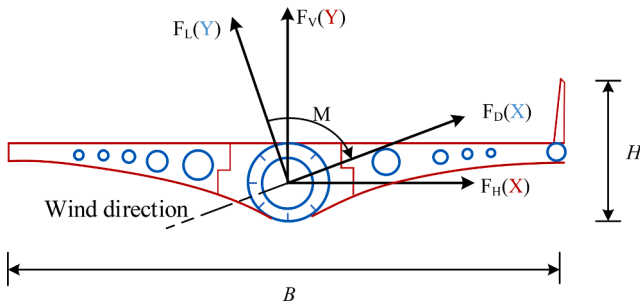


Fig. 13. Definition of wind load in different coordinate systems.

$$\text{Lift : } C_L = \frac{F_L}{\frac{1}{2}\rho U_\infty^2 B} \quad (7-2)$$

$$\text{Moment : } C_M = \frac{M}{\frac{1}{2}\rho U_\infty^2 B^2} \quad (7-3)$$

where U represents the mean wind speed, which is equal to 7.02 m/s; ρ is the air density, equal to 1.225 kg/m³; B is the girder's width of segment model, $B = 0.1$ m; H is also the whole height of main girder and railing for the model, $H = 0.041$ m, as illustrated in Fig. 13; l is the length of the segment model, as shown in Fig. 12; F_D , F_L , and M are the wind loads acting on the main girder in the wind coordinate system; F_H and F_V are respectively the drag force and lift force in the body coordinate system; the moment coefficients are the same in the two coordinate systems.

These tests aim to measure the drag (F_D), lift (F_L), and torsional moment (M) of the deck per unit span for the four geometries considering the effect of the main girder segment on the experimental results; a comparison of the force coefficients on the main girder in a uniform flow is illustrated in Fig. 14. An examination of the experimental results and a comparison of the aerodynamic coefficient C_D in Fig. 14 (a–c) suggest that MOD1, MOD2, and MOD3 exhibit similar aerodynamic behaviours, slightly different from that exhibited by MOD4. On average, the C_D of MOD4 is smaller than those of MOD1, MOD2, and MOD3 in the interval $-12^\circ \leq \alpha \leq 0^\circ$, whereas in the interval $-2^\circ \leq \alpha \leq 12^\circ$, MOD1 has the smallest C_L and C_M values. Regarding C_M , MOD3 and MOD4 have the smallest values, whereas MOD1 and MOD2 have the largest ones. At the 12° wind angle of attack, the drag force coefficients for the four models are respectively 2.06, 2.136, 2.001, and 2.002, while those values at the -12° wind angle of attack are 2.001, 2.102, 2.001, and 1.737. The drag force coefficient at the 12° wind angle of attack is about 97% that at the -12° wind angle of attack. Both the lift and moment coefficients decrease with an increase of the wind angle of attack, but the drag coefficients exhibit a slight variation within the range of -12° to 12° . The

above results indicate that the aerodynamic forces vary slightly along the bridge longitudinal axis, and the lift and moment coefficients are more sensitive to the wind angle of attack than the drag coefficient.

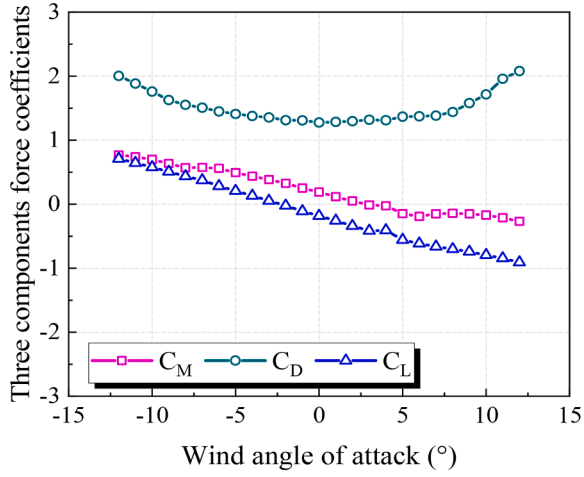
5. Results and discussion

In this section, the test results in terms of the VIV for the bridge were presented and analysed; subsequently, this paper investigated the effect of damping ratio on the aerodynamic response of the bridge. The aerodynamic improvement measure was then studied to enhance the aerodynamic stability by retrofitting railings. The wind speed at the girder level is considered as the reference value in the analysis. All the following results and wind speeds are presented at the prototype scale.

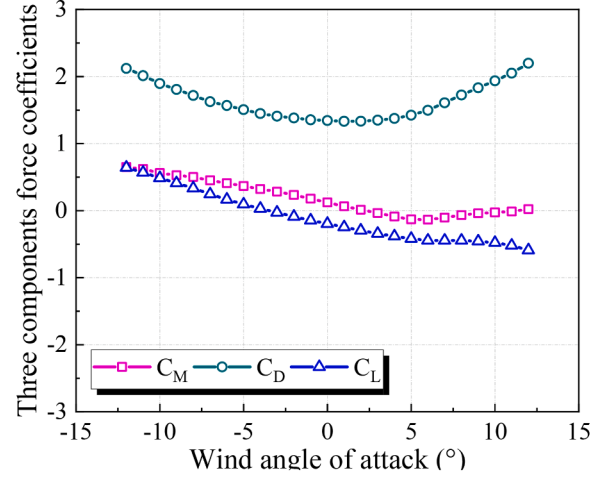
5.1. Vortex-induced vibration analysis

VIV generally tends to affect the fatigue damage to crucial structural components and bridges life. To observe the VIV within the locked-in wind speed for the bridge deck in a uniform flow field. Fig. 15 depicts the vertical displacement and torsional angle at the mid-span section of main girder. The vertical displacements at measuring Points 9 and 10 were taken into account for the vertical displacement root-mean-squares (RMS) calculation using the commercial software of MATLAB. Fig. 16 shows the vertical displacement and lateral displacement RMS at the end of the main girder. The displacement of measuring Points 7 and 6 were respectively considered to calculate the corresponding displacement response RMS. The arrangement of the corresponding measuring points is shown in Fig. 5(b). Fig. 17 depicts the power spectrum of wind-induced vibration response at mid-span and end of main girder.

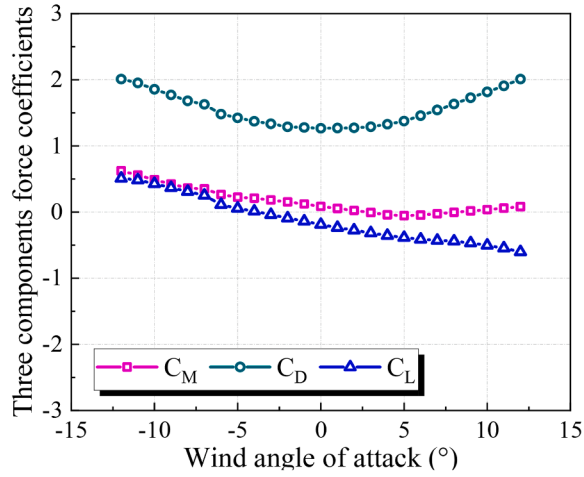
As shown in Fig. 15, the mid-span section of the main girder exhibits vertical and torsional VIVs at the -3° , 0° , and $+3^\circ$ wind angles of attack, in particular the drag tends to the minimum at 0° wind angle of attack (see in Fig. 13). The vertical vibration RMS at the mid-span section are, respectively, 12.7 mm, 21.1 mm, and 31.9 mm, and the corresponding locked-in wind speeds are [3.28 m/s, 8.37 m/s], [22.56 m/s, 26.93 m/s], and [23.91 m/s, 29.29 m/s]. The torsional VIVs at the mid-span section of the main girder are respectively 0.302° and 0.162° corresponding to the locked-in wind speed of [11.95 m/s, 17.99 m/s]. Note that the torsional VIV also occur at the wind speed of 41.32 m/s corresponding to the amplitude of 0.170° . The researches results illustrate that the VIV not only occurs in a low wind speed range of 6 m/s–27 m/s, but also in the high wind speed of 41.32 m/s. The reason is that the vortex shedding alternately forms on both sides of the mid-span section of main girder, leading to the vortex excitation force acting on the structure, and the cross-section of the main girder exhibits VIV under the harmonic force. On the other hand, the synchronous dynamic load caused by the fluctuating wind may act on various locations of the



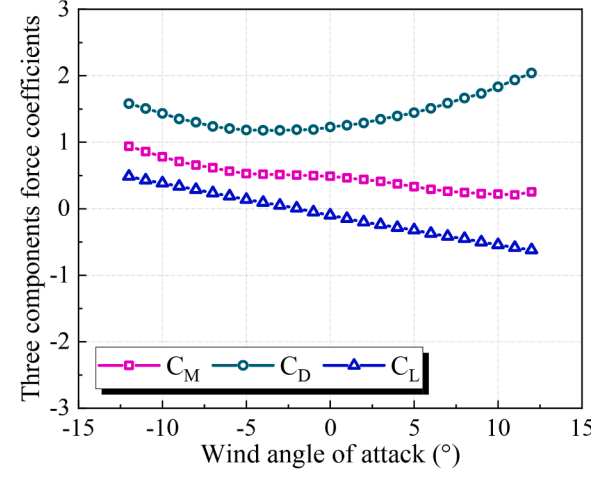
(a) MOD 1



(b) MOD 2

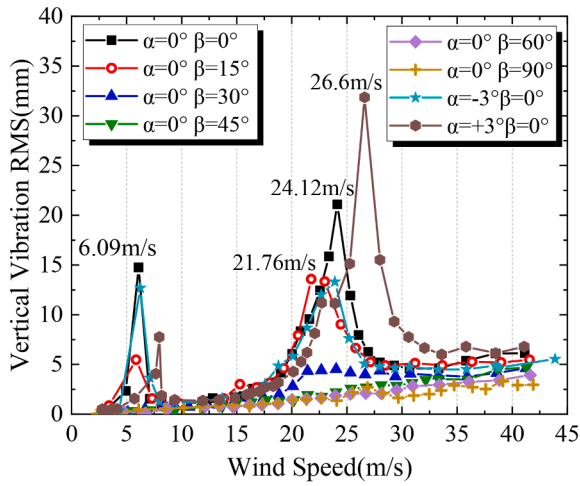


(c) MOD 3

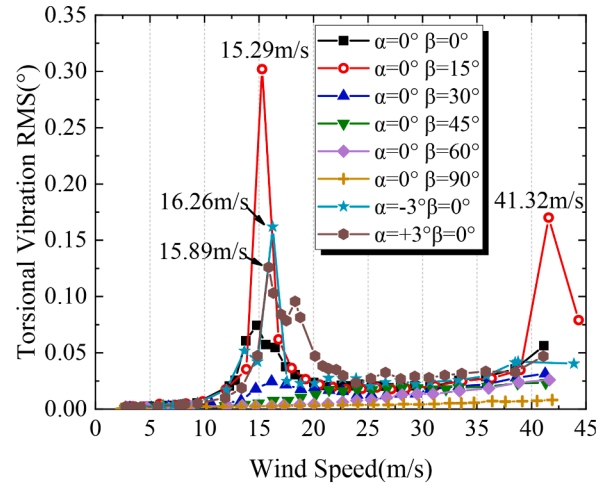


(d) MOD 4

Fig. 14. Three-component force coefficients for the four conditions.



(a)



(b)

Fig. 15. Vertical and torsional vibration RMS at the mid-span section of the bridge.

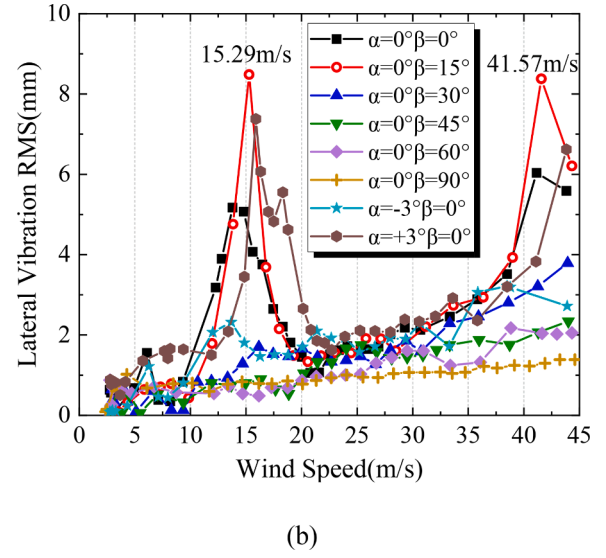
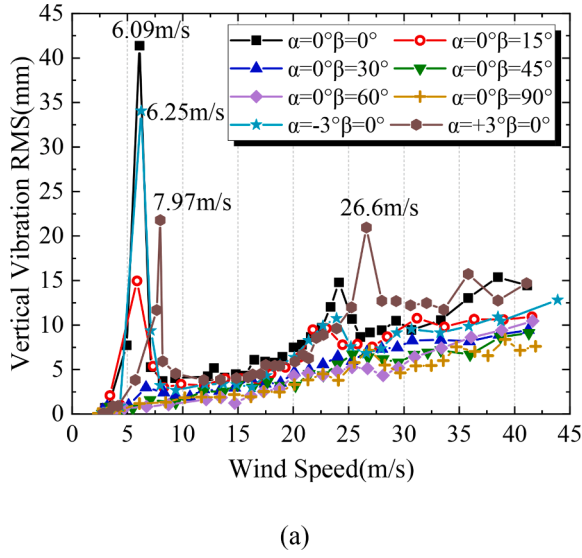


Fig. 16. Vertical vibration RMS and lateral displacement RMS at the end section of the bridge.

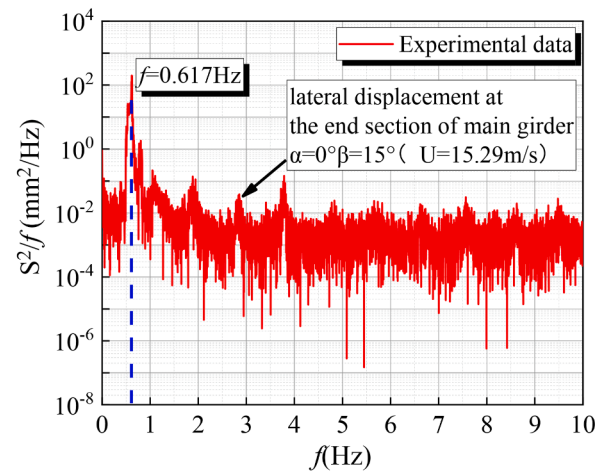
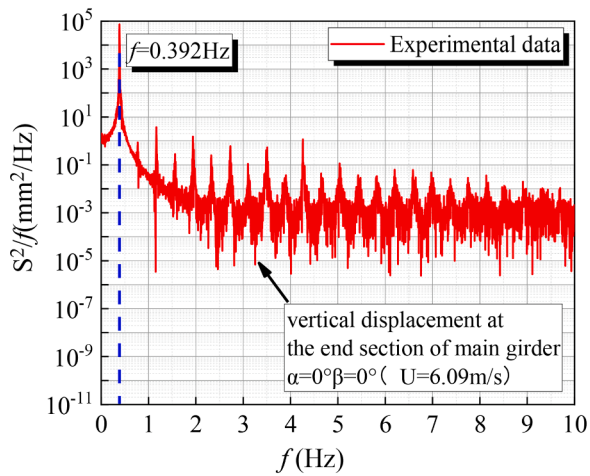
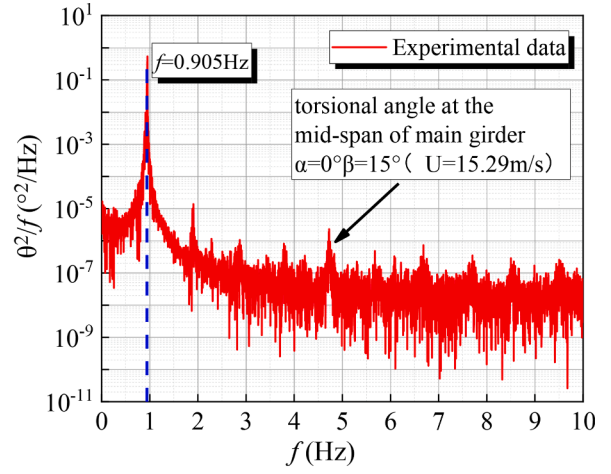
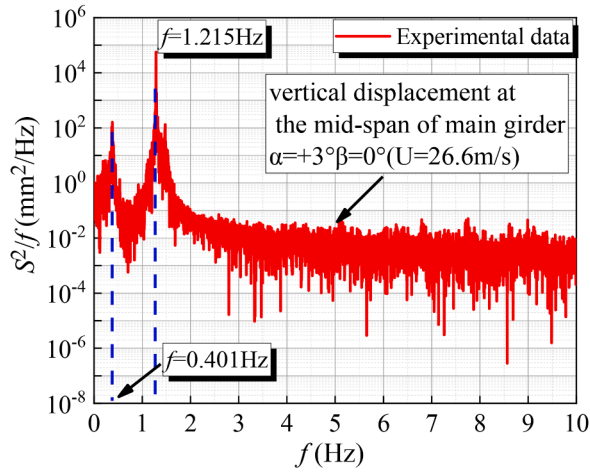


Fig. 17. Power spectrum of wind-induced vibration response.

bridge owing to the large turbulence integral scale within the wind speed ranging from 6 m/s to 42 m/s (see in Fig. 11), resulting in VIV for the bridge.

At the end section of the main girder, the vertical vibration RMS at the -3° , 0° , and $+3^\circ$ wind angles of attack are 41.37 mm, 34.07 mm, and 21.79 mm, corresponding to the locked-in wind speed of [3.42 m/s, 9.37 m/s]. The wind speed caused the vertical VIV is <10 m/s, but the amplitude of vortex-induced response (VIR) is higher than that of the mid-span of main girder. Based on the test statistics, VIV is ascribed a restricted vibration, and the amplitudes of the first two vertical bending vortex-induced resonances are 109.29 mm and 33.90 mm, respectively, and those of torsional vortex-induced resonance are 0.198° and 0.155° .

Obviously, the structure responses become a single-frequency amplitude, which match with the harmonic signal. It is found that a two-mode vibration with higher peaks when the frequency of vortex-shedding (0.401 Hz and 1.215 Hz) respectively matches with the first-order and second-order flexural vibration frequency (0.366 Hz and 1.180 Hz) of the bridge (see in Fig. 17(a)). Consequently, the two-mode vertical VIV generated at the mid-span section of main girder under the wind velocity of 26.6 m/s. The torsional angle power spectrum presents a sharp peak at the torsional vibration frequency (0.905 Hz) with reference to a low wind speed, and the frequency is close to the first-order torsional vibration frequency 0.921 Hz, resulting in the torsional VIV, as shown in Fig. 17(b). From Fig. 17(c), the vibration for bridge is approximately a single-mode vibration and a higher peak is visible at a frequency of 0.392 Hz, while the other peaks are all small. It is observed from Fig. 17(d) that a higher peak at the frequency of 0.617 Hz generates with reference to the wind velocity of 15.29 m/s, leading to the lateral VIV.

The influence of turbulence on the dynamic displacements of the CSBB is investigated (Figs. 18 and 19) and the turbulent wind field is simulated by equipping roughness elements and spires in the front part of the model (illustrated in Fig. 10). The corresponding wind characteristics in the wind tunnel are equivalent to the type D stipulated in the JTG/T3360-01-2018 code [22], as shown in Fig. 10.

Fig. 18 depicts the vertical vibration responses of the bridge girder in the completed state to investigate the influence of the flow field. It is evident that the vertical VIV occurs in the uniform flow, and a sharp peak at 1.178 Hz is clearly identified that closes to the second-order vertical flexural vibration frequency 1.180 Hz (see Fig. 19(a)). However, there is no explicit VIV in the turbulent flow (see Fig. 18(b)), and some small peaks distributed around 5 Hz is visible at the identical wind velocity. The bridge deck seems to experience vortex shedding, and the

bridge deck behaves the lower vibrations because turbulence can impair the correlation of vortices along the bridge span and the variation in aerodynamic forces. The results indicate that the aerodynamic performance of the bridge in the turbulent flow field is better than that in the uniform flow field.

Table 6 summarises the first two vertical and torsional VIVs within the wind speed range of 0.0 m/s–58.4 m/s in a uniform flow. The damping ratio of the bridge obtained in the wind tunnel test is 0.73%. It is notable that the vertical and torsional VIVs occur at 0° and $+3^\circ$ wind angle of attack, and only torsional VIV occurs at the -3° wind angle of attack. This is because the cross-section of the main girder is a blunt body at a positive wind angle of attack corresponding to along the bridge girder axis, which will cause a separation flow.

At the -3° wind angle of attack, the cross-section is relatively more streamlined, and thus, there is no explicit vertical VIV. Obviously, considering the effects of wind speed and wind yaw angle, the vertical VIV generates at the lock-in wind speeds of 23 m/s and 41 m/s when the wind yaw angles are 0° and 15° . The torsional VIV occurs at the 23 m/s wind speed when the wind angle of attack and yaw angle are also respectively 0° and 15° . The results demonstrate that the main girder at positive wind angles of attack and the relatively larger wind speeds is more prone to VIV.

The Strouhal number is an important factor to evaluate the aerodynamic characteristics of a bluff body, especially in estimating the critical wind velocity of VIV and the amplitude of the VIV, expressed as Eq. (8) [25]. The Strouhal number locates within the range of 0.080–0.614 (listed in Table 6), and the amplitude of vertical and torsional VIV at $+3^\circ$ and 0° wind angle of attack are much larger than the corresponding limits.

$$St = \frac{f_v B}{U} \quad (8)$$

where B is the bridge's characteristic size perpendicular to the incoming flow, which ranges from 5 m to 10 m; U is the corresponding locked-in wind speed; f_v is the vortex shedding frequency.

5.2. Effects of damping ratio and aerodynamic configuration

Table 7 summarises the effect of the damping ratio on the wind-induced response amplitude at 0° wind angle of attack and wind yaw angle. In the wind tunnel test, a high-energy dissipation viscoelastic material, that is, butyl rubber pads with a plane size of $A_0 = 70 \times 50$ mm and a thickness of $h_0 = 10$ mm is chemically bonded to the lower surface

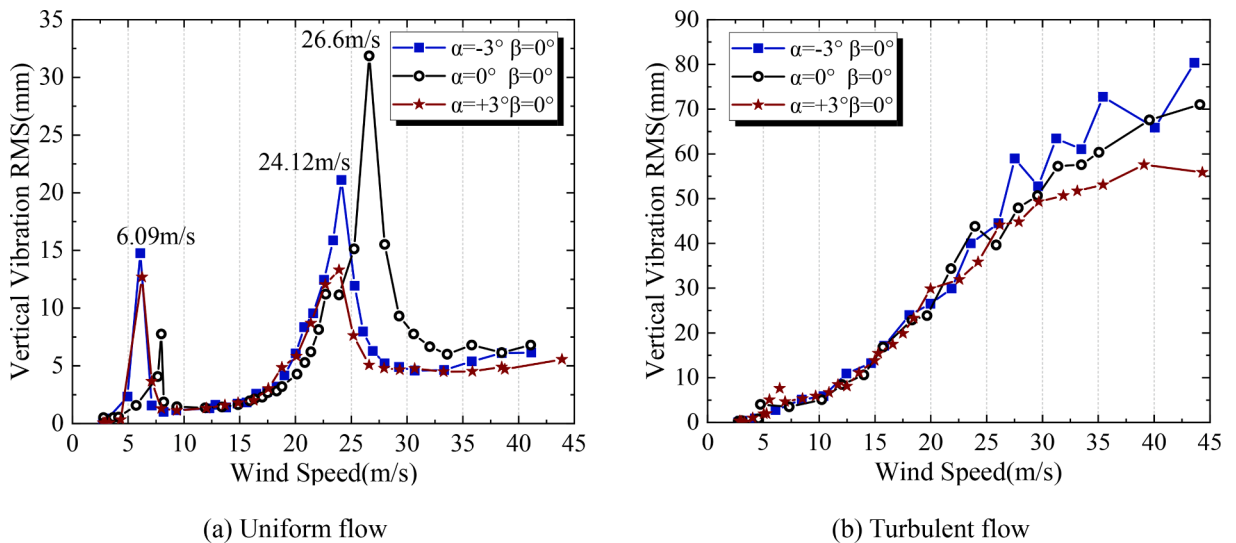


Fig. 18. Wind-induced response at the mid-span section of the main girder.

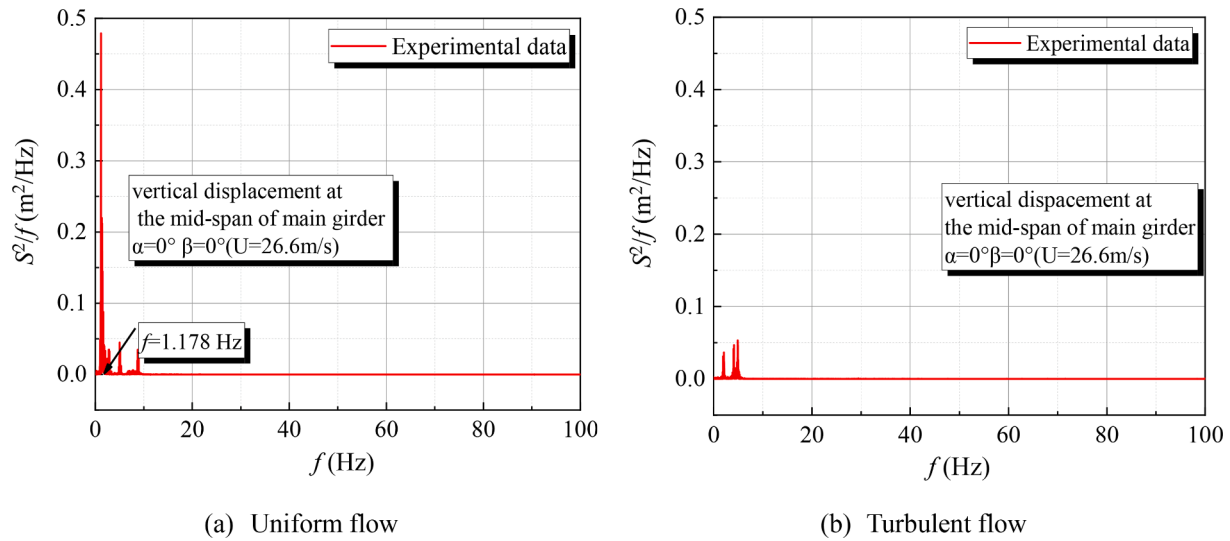


Fig. 19. Power spectrum of vertical displacement.

Table 6

Obvious vortex vibration in the uniform flow.

$\alpha(^{\circ})$	$\beta(^{\circ})$	Wind speed (m/s)	Torsion/Vertical	Frequency (Hz)	Amplitude (mm/ $^{\circ}$)	S_t	VIV
0	0	6	Vertical	0.366	105.0 < 109.29	/	N
	0	23	Vertical	0.366	38.14 > 33.9	0.080–0.159	Y
	0	15	Torsion	0.921	0.160 < 0.198	/	N
	0	41	Torsion	0.921	0.121 < 0.155	/	N
	15	6	Vertical	0.366	37.88 < 109.29	/	N
	15	23	Vertical	0.366	24.59 < 33.9	/	N
	15	15	Torsion	0.921	0.302 > 0.198	0.307–0.614	Y
	15	41	Torsion	0.921	0.170 > 0.155	0.112–0.225	Y
+3	0	8	Vertical	0.366	55.14 < 109.29	0.070–0.141	N
	0	26	Vertical	0.366	57.60 > 33.9		Y
	0	16	Torsion	0.921	0.272 > 0.198		Y
−3	0	6	Vertical	0.366	88.26 < 109.29	0.288–0.576	N
	0	23	Vertical	0.366	24.07 < 33.9		N
	0	16	Torsion	0.921	0.162 > 0.155		Y

Note: α is the wind angle of attack, β is the wind yaw angle. N: there is not VIV. Y represents VIV.

Table 7

Effect of damping ratio on wind-induced response amplitude.

	$\alpha(^{\circ})$	$\beta(^{\circ})$	Wind speed (m/s)	Torsional/Vertical	Amplitude (mm/ $^{\circ}$)		Reduced percentage (%)	Limit
					$\zeta = 0.005$	$\zeta = 0.012$		
Smooth flow	0	0	6	Vertical	105.00	31.46	70	conform
		0	23	Vertical	38.14	20.41	46	conform
		0	15	Torsional	0.160	0.105	34	conform
		0	41	Torsional	0.121	0.091	25	conform

of the main girder to simulate the effect of damping ratio on the amplitude.

At the damping ratio 0.5%, the vertical displacement of the main girder is about 105.5 mm at the 6 m/s wind speed. In contrast, the displacement is 31.46 mm as the damping ratio increases to 1.2%; thus, the displacement of the bridge girder at 1.2% damping ratio is about 30% that at 0.5% damping ratio. The vertical displacement of the main girder at the damping ratio 1.2% is about 53% that at the 0.5% damping ratio with the loading of the 23 m/s wind speed. The torsional displacement of the main girder is respectively 0.16° and 0.121° at the damping ratios of 0.5% and 1.2%. The torsional displacement of the main girder at the damping ratio 1.2% is about 75% of that at the 0.5% damping ratio at the 41 m/s wind speed. Overall, the damping can effectively reduce the wind-induced response amplitude to improve the aerodynamic stability of the structure.

Fig. 20 illustrates the RMS of the vertical and torsional VIR of the CSBB with airtight railing and ventilated railing (see in Fig. 6) at 0° wind angle of attack and wind yaw angle. For the airtight railing, the vertical vibration RMSs at the mid-span section are 7.24 mm and 17.2 mm and the corresponding locked-in wind speeds are [3.28 m/s, 7.59 m/s] and [19.82 m/s, 26.49 m/s]. Compared with the airtight railing, the vertical vibration amplitude for the ventilated railing is reduced by about 72%. The torsional vibration RMS in the mid-span section is 0.07°, corresponding to the locked-in wind speed of [10.5 m/s, 18.52 m/s]. More notably, the torsion amplitude is reduced by about 86% when the airtight railing is instead by ventilated railing.

Note that no VIV is observed for the bridge when the railing is designed as the ventilated railing. Also, the responses are significantly reduced within the range of 20 m/s–45 m/s, which is in accordance with the phenomena that the vertical VIV is remarkably mitigated for the

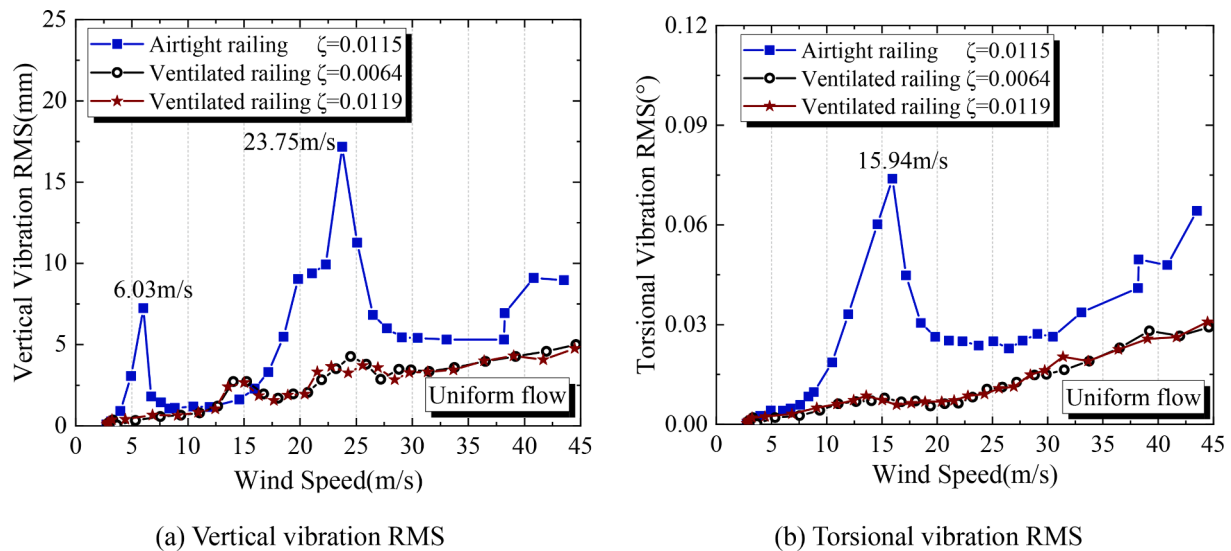


Fig. 20. Vortex resonance of bridge at the mid-span section.

twin-box girders by installing the attachments reported by Laima et al. (2018) [32]. A significant improvement in the aerodynamic stability of the bridge is testified by enhancing the structure aerodynamic configurations, indicating that VIV is sensitive to the aerodynamic configuration.

6. Conclusions

This study investigates the aerodynamic performance of a CSBB in a uniform and turbulent flow through a scaled full-aeroelastic model wind tunnel test and segmental model test. The following conclusions are drawn within the research scope, which can provide engineering reference for wind resistance and vibration reduction design of this kind of bridge.

- (1) The bridge deck is prone to the vertical vortex-induced resonance in a low turbulence intensity within the wind speed ranging from 6.09 m/s to 26.6 m/s as the wind direction nearly perpendicular to the longitudinal bridge axis (at the 0° and 15° wind yaw angles), but the maximum wind speed causing the torsional vortex-induced vibrations increases to 41.32 m/s for the light bridge.
- (2) The responses for the bridge deck present a sharp single-frequency amplitude when the frequency of vortex-shedding respectively matches with the flexural vibration frequency in the uniform flow field whereas only some small peaks are visible in the turbulent flow.
- (3) The vortex-induced vibrations disappear in turbulence flow because turbulence can impair the correlation of vortices along the bridge span. The damping can effectively alleviate the wind-induced response amplitude, note that it is reduced by 70% when the damping ratio increases to 1.2%.
- (4) The ventilated railings exert significant suppression effects on the vertical and torsional vortex-induced vibration, also the wind-induced vibration amplitudes are remarkably mitigated with adopting the ventilated railings, which is of great promise in engineering application.

CRediT authorship contribution statement

Qing Sun as the corresponding author, is mainly responsible for proposing conceptualization, project administration, and funding acquisition.

Yanru Wu as the first author, performed the data curation, formal

analysis, and interpretation of data for the work, and then written the original draft and revised it critically for important intellectual content.

The other authors participated in methodology, formal analysis, and provided editing and writing assistance to the manuscript, who were ranked based on their contributions

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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